Elections on measured stakes: welfare tracking and platform competition with microsimulated household impacts

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Abstract

Does plurality voting select the policy a stated social welfare function prefers when voters misperceive their own stakes? I study the question in a simulation whose inputs are measured and whose behavioral layers are explicit modeling choices: the electorate is 120,408 voting-age adults from certified, population-calibrated US microdata; the two platforms are actual encoded tax reforms with opposite incidence, budget-matched within 3%; each voter's stake is the tax engine's computed change in their household's net income, closed by a labeled financing rule; and perception — truth plus Gaussian noise plus optional shared bias — is the headline assumption. For plurality the model admits an analytic normal approximation at any electorate size, exact in the large-population limit. Measured stakes are two-cliffed: net of financing, three-quarters of adults hold under \$25 a year while over a fifth hold more than \$500, and this structure overturns standard toy-model intuitions. Tracking is non-monotone in accuracy: at perfect information the outcome is decided by the *sign* of a \$4.10-per-adult financing residual — a welfare-irrelevant quantity — while moderate noise ($\sigma \approx$ \$50–\$2,000) restores near-certain tracking by turning the trivial-stakes mass into canceling coin flips. Tracking thresholds are electorate-size statements that collapse toward 0.5 accuracy as n grows, while the expected-vote-share crossing behind the failure ceiling does not move with n. And shared misperception of $\approx$ \$221 per voter per year flips elections that \$1,000 of independent noise cannot move. Making the platforms choices reverses the information verdict: when two candidates — households from the data, each valuing their own stake and society's equally-distributed-equivalent income change on one dollar scale — pick positions in the space the measured incidence vectors span, perfect information becomes an undercutting force that drives every pure Nash equilibrium to the status quo, the welfare optimum available, while noise lets both self-serving programs escalate to full intensity and erodes the electoral advantage of the broad-constituency program over the narrow one. Giving voters the same mixed motives restores order cheaply: 3.8% of utility weight on society cures the knife-edge, a 10% informed-sociotropic minority disciplines purely selfish candidates back to the status quo at every noise level (traversing a cycling regime around 2–5% shares where the pure-equilibrium set can be empty), and the residual distortion is value pluralism — fully informed sociotropic voters split when their inequality aversions disagree. The mechanisms are classical; the contribution is dollar-denominated calibration on measured microdata, a reusable open-source harness, and a direct path to replacing the perception assumption with survey estimates.

*max@maxghenis.com. Code, data artifact, and every number's regeneration script: `https://github.com/MaxGhenis/democrasim`. The author co-founded PolicyEngine, whose open models and data this personal project uses.

1 Introduction

A large literature asks when majority voting aggregates dispersed information or dispersed interests into good collective choices. The Condorcet jury theorem gives the optimistic benchmark: homogeneous stakes and independent errors let even barely-informed majorities decide almost perfectly (Condorcet, 1785). Its known failure modes are equally classical: correlated errors break the aggregation (Ladha, 1992), the crux of the miracle-of-aggregation debate (Page and Shapiro, 1992; Caplan, 2007); intense minorities lose to indifferent majorities under one-person-one-vote (Dahl, 1956); rationally ignorant voters may economize on information (Downs, 1957), and less-informed voters may strategically abstain in common-value settings (Feddersen and Pesendorfer, 1996) — a distinct mechanism from the private-stake indifference abstention modeled here; and probabilistic-voting models tie what elections aggregate to the responsiveness of vote probabilities (Coughlin and Nitzan, 1981; Lindbeck and Weibull, 1987).

What this paper adds is not a new mechanism but a new input quality. Simulation exercises in this tradition typically invent the key quantity — the distribution of what voters stand to gain or lose — and the invented distributions drive the conclusions. Here the inputs are measured and the behavioral layers are explicitly labeled. The electorate is population-calibrated microdata; the platforms are actual encoded reforms; the stakes are computed by a tax-benefit microsimulation engine, household by household, including interactions across programs. On these inputs, several toy-model conclusions reverse, and the surviving ones acquire dollar denominations: how many dollars of noise elections tolerate (thousands), how many dollars of shared bias they do not (about \$221 per voter per year), and how many dollars of stake the decisive voting bloc holds at perfect information (\$4.10 per adult per year, net of financing).

Five fixed-platform results come first. First, measured stakes are nothing like the homogeneous or Gaussian stakes of standard toys: 76% of adults have approximately \$0 at stake gross of financing, while a fifth hold stakes above \$500 (Section 4.1). Second, welfare tracking is non-monotone in perception accuracy, and its failure at perfect information is a knife-edge with a precise anatomy: the outcome becomes *welfare-independent*, set by the sign of the residual cost gap between the two policies (Section 4.2). Third, every tracking threshold is an electorate-size statement; a closed form places the Monte Carlo results on the full ladder from 101 voters to the large-population limit (Section 4.3). Fourth, elections in this environment are robust to large independent misperception and fragile to small shared misperception (Section 4.4). Fifth, moment-matched Gaussian comparators — even preserving impact–income covariance — reproduce none of this structure, so calibrating stake *shape* matters more than matching moments (Section 4.5).

Two further layers then make behavior endogenous, on both sides of the ballot. Platforms become choices: two policy-motivated candidates (Wittman, 1983; Calvert, 1985) — households drawn from the data, in the citizen-candidate spirit (Osborne and Slivinski, 1996; Besley and Coate, 1997) — pick positions in the two-dimensional space the measured incidence vectors span, and every pure Nash equilibrium is computed exactly (Section 5). And voters get the same utility family the candidates carry: a selfish weight on their own perceived stake and a societal weight on the policy's equally-distributed-equivalent income change (Atkinson, 1970), the sociotropic motive of Kinder and Kiewiet (1981) and the altruistic-voting calculus of Edlin et al. (2007) in dollar form, with mixtures of types heterogeneous in weights, inequality aversion, and information quality (Section 6). The two layers interact: the information conditions under which elections grade fixed policies well are close to the opposite of those under which they discipline the policies candidates offer.

The exercise is a thought experiment about one mechanism: self-interested voting on misperceived, engine-computed household impacts. It is not political science and predicts no elections; real voters weigh values, identity, and information far beyond their own net income (Bartels, 2005; Stantcheva, 2021). Platforms are labeled generically throughout.

2 Model

Electorate. Each voter i is a voting-age adult carrying their household's true impact δ_{ij} (dollars per year) under each policy $j \in \{A, B\}$, a survey weight, baseline household net income, and the household's adult count a_i. Voting is per adult; welfare counts each household's dollars once (contributions divide by a_i).

Financing. Engine impacts are gross: a tax cut shows only winners. The headline specification levies each policy's total cost equally per adult, making both policies budget-neutral by construction; a proportional-to-income levy and the unfinanced gross case are stress tests. Net stakes and the margin $m_i = \delta_{iA} - \delta_{iB}$ are defined after financing.

Welfare. An isoelastic social welfare function over household net income ($\eta = 1$, incomes floored at \$1,000) ranks the two policies; "tracked" means the election selects the *ballot-best* policy, the higher-ranked of the two. Under this metric both financed policies score below the status quo, which plurality does not offer; Section 4.2 returns to this.

Perception and voting. Voter i perceives $\hat{\delta}_{ij} = \delta_{ij} + b_j + \varepsilon_{ij}$ with $\varepsilon_{ij} \sim N(0, \sigma^2)$ i.i.d. across voters and policies, and b_j a shared bias. Under plurality with abstention at exact perceived indifference, voter i votes for A with probability

$$p_i(\sigma, b) \;=\; \Phi\!\left(\frac{m_i - (b_B - b_A)}{\sigma\sqrt{2}}\right) \qquad (\sigma > 0), \tag{1}$$

while at $\sigma = 0$ the vote is deterministic: A if the shifted margin is positive, B if negative, abstention if exactly zero. The implementation also supports multiplicative attenuation of m_i, fixed to 1 throughout this paper. so an election of n voters sampled with probability proportional to weight reduces to a multinomial over (A, B, abstain) with population-weighted mean probabilities $(\bar{p}_A, \bar{p}_B, \bar{p}_0)$. With $g = \bar{p}_A - \bar{p}_B$, the probability that A wins is $\Phi\big(ng/\sqrt{n(\bar{p}_A + \bar{p}_B - g^2)}\big)$ by the normal approximation, degenerating to a step function of g as $n \to \infty$. All plurality results use this normal approximation (exact in the large-population limit) or Monte Carlo simulation that matches it; approval voting and instant-runoff variants are simulated directly, and with two policies instant-runoff coincides with plurality absent exact ties.

3 Data and calibration

The electorate comes from PolicyEngine's certified, population-calibrated Populace microdata for the United States (57,240 households; 120,408 adults representing 268 million), scored with `policyengine-us` 1.764.6 for policy year 2026. Policy A raises the federal Child Tax Credit base amount from \$2,200 to \$3,200; Policy B caps the top two marginal income tax rates at

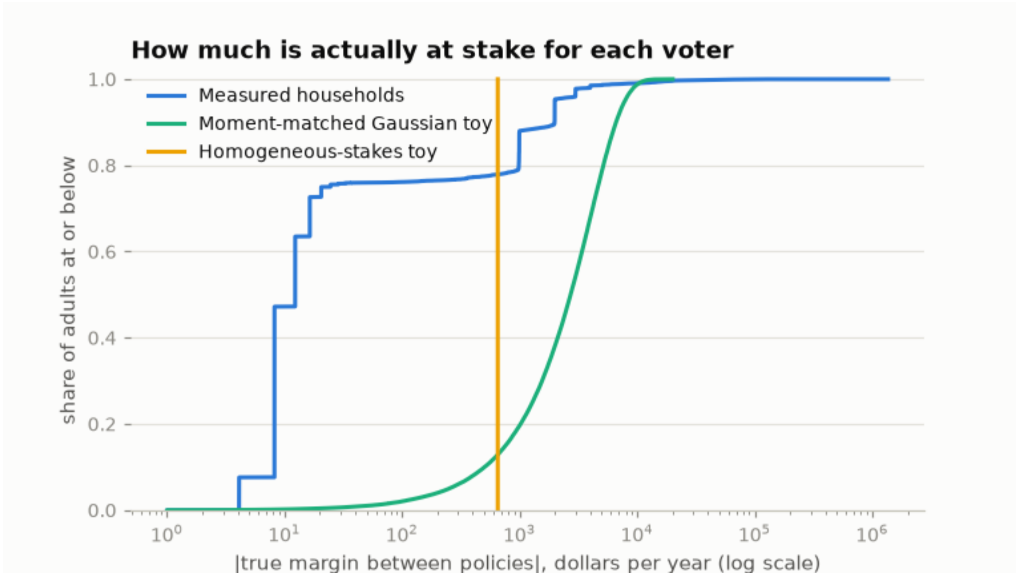

Figure 1: Weighted cumulative distribution of $|m_i|$, the true margin between the two policies per adult, net of per-capita financing. Measured households form a staircase — 47% of adults within \$10, 75.5% within \$25, driven by the \$4.10-per-adult difference in the two policies' levies — with a high-stakes tail. The two synthetic comparators used by toy models lack this structure.

34%. The engine scores them at \$39.1B and \$38.0B a year of household net income change (model totals including state-tax interactions, not budget scores) — within 2.9%, so neither platform is simply the larger transfer. Incidence is opposite: 9.9% of Policy A's dollars reach the top income decile versus 99.7% of Policy B's.

The committed data artifact carries full provenance (engine versions, certified dataset revision and hash, reform parameter dictionaries) and passes builder validations that reconcile every extracted quantity against the engine's own weighted aggregates (worst relative discrepancy 10^{-16}). Regression tests pin the artifact facts the results below depend on — most fragilely, the sign of the cost gap — so a future data rebuild cannot silently invert them.

4 Results

4.1 Measured stakes are two-cliffed

Gross of financing, 76% of adults see approximately \$0 from both policies; Policy A's gains arrive in per-child quanta of about \$1,000 for the 20% of households with eligible children. Financing hands the untouched majority a small signed stake: the two levies differ by \$4.10 per adult per year, so 75.5% of adults hold margins under \$25 while 19.9% (all in households with children) hold pro-A margins above \$500 and 2.6% hold pro-B margins below $-\$500$ (Figure 1). The weighted mean absolute margin, \$652, describes almost nobody.

4.2 Non-monotone tracking and the welfare-independence knife-edge

At 10,001 sampled voters, tracking is essentially perfect for noise σ between about \$50 and \$2,000 (every estimate 1.000, Wilson lower bounds 0.984) and collapses at both extremes

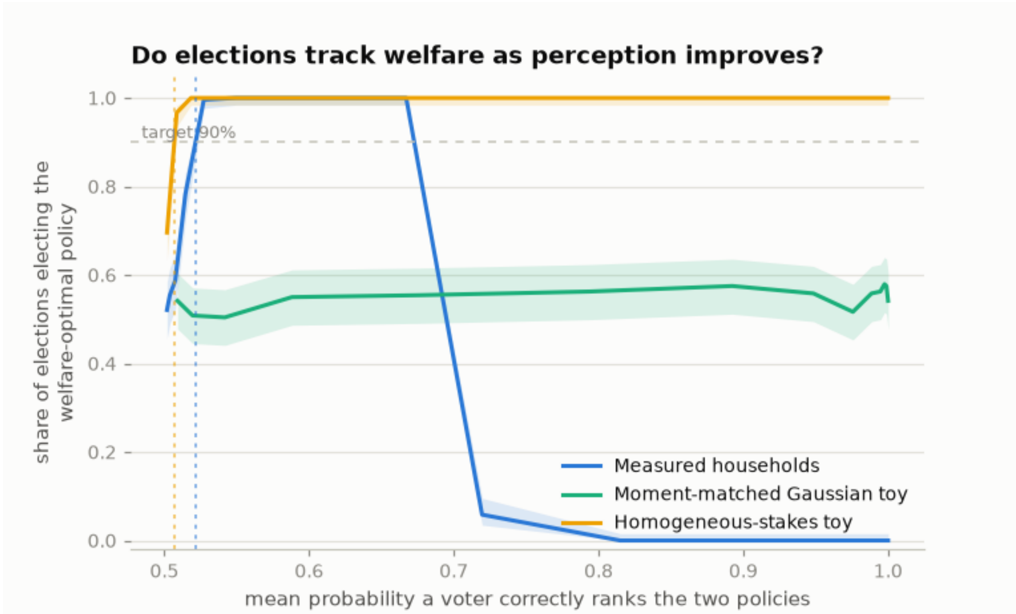

Figure 2: Share of 10,001-voter elections electing the ballot-best policy (240 elections per point; Wilson 95% bands), against mean ranking accuracy — the probability a voter correctly ranks the two policies for their own household. The homogeneous toy is the Condorcet jury theorem; the measured curve fails at *both* ends.

(Figure 2). The high-noise decay is ordinary. The perfect-information failure is not: at $\sigma = 0$, 78.7% of adults strictly prefer Policy B — including the 76% of all adults untouched by either reform, whose entire stake is the \$4.10 levy gap — and plurality elects B in every election, the opposite of the welfare ranking.

Table 1 dissects the knife-edge. The direction is set entirely by the sign of the residual cost gap: rescaling Policy B to cost 3% *more* than A flips perfect-information tracking from 0% to 100% with the welfare ranking unchanged. The mass only rules while it votes: exact parity or unfinanced gross stakes leave it exactly indifferent, it abstains (turnout 24%), and tracking holds from accuracy ≈ 0.59 through perfect information. A behavioral \$25 indifference band does the same; a \$10 band does not, because multi-adult households' levy margins exceed it. Approval voting reaches the same outcome through self-silencing — voters who perceive both policies as net losses approve neither — though under the three-option ranking the welfare metric itself induces (status quo first), a winner elected on 23% turnout is still a welfare-reducing enactment; approval's property here is the filter, not welfare optimality.

Perfect information therefore does not make elections anti-track welfare; it makes them *welfare-independent*, delegating the outcome to whichever side of an arbitrary residual the no-stakes bloc shares. Between the extremes, moderate misperception helps: noise converts four-dollar margins into fair coins that cancel, and the informed high-stakes minority decides. This parallels the logic of probabilistic-voting models (Coughlin and Nitzan, 1981; Lindbeck and Weibull, 1987): idiosyncratic noise makes plurality weight stakes, and vanishing noise reverts it to counting heads — here traversed numerically on measured stakes.

Variant (perfect information)	Tracked	Turnout	Decisive margin
Per-capita financing (headline)	0%	100%	−$4.10 levy gap
Proportional financing	0%	99.8%	same sign, income-scaled
Sign-flip counterfactual (B costs +3%)	100%	100%	+$4.38 levy gap
Exact-parity counterfactual	100%	24%	$0 (mass abstains)
Gross stakes (no financing)	100%	24%	$0 (mass abstains)
Abstention band $10	0%	53%	multi-adult margins survive
Abstention band $25	100%	24%	staircase silenced
Approval voting	100%	23%	mass approves neither

Table 1: The perfect-information knife-edge under stress. Counterfactual rows are synthetic rescalings of Policy B's gross impacts, labeled as such in the regeneration scripts.

4.3 Thresholds are electorate-size statements

The accuracy band in which tracking exceeds 90% runs from 0.598–0.641 at $n = 101$ to 0.502–0.689 in the large-population limit (Figure 3). The entry threshold is Condorcet arithmetic and collapses toward 0.5 as n grows. The ceiling is pinned by an expected-vote-share crossing that does not move with n; the 90% boundary reaches it (≈ 0.69) from $n = 1{,}001$ onward, with only the smallest electorate (0.641 at $n = 101$) held below it by sampling noise. Any reported "accuracy threshold for elections to work" that does not state its electorate size is a sample-size artifact.

4.4 Bias, not noise, is the cheap attack

Holding $\sigma = \$1{,}000$ — where unbiased elections track essentially always — a shared misperception that Policy B is worth more than it is flips the median 10,001-voter election at $\approx \$221$ per voter per year (1.7% of elections flip at $178, 97.5% at $267). That is one-third of the mean absolute stake and one-fifth of a single per-child credit step. The contrast instantiates the correlated-errors breakdown of jury aggregation (Ladha, 1992) and the miracle-of-aggregation debate (Page and Shapiro, 1992; Caplan, 2007) in dollars: independent error in the thousands is harmless; shared error in the low hundreds is decisive.

4.5 Moment-matched toys reproduce none of it

A Gaussian electorate matched to the measured means and covariances of (impact_A, impact_B, log income) — preserving the incidence correlation that carries the welfare signal — has smooth, nearly centered margins: its perfect-information vote margin is a fraction of a percentage point. At 10,001 voters it never reaches 90% tracking; in the large-population limit it tracks with near-certainty. Neither behavior resembles the measured world's wide certainty band bounded by a knife-edge. The homogeneous-stakes toy (every voter holds the mean absolute margin) is the jury theorem, tracking from accuracy 0.507 at this n. What separates the measured world from both is not its first or second moments but its shape: the coexistence of a nearly indifferent majority with a concentrated high-stakes minority.

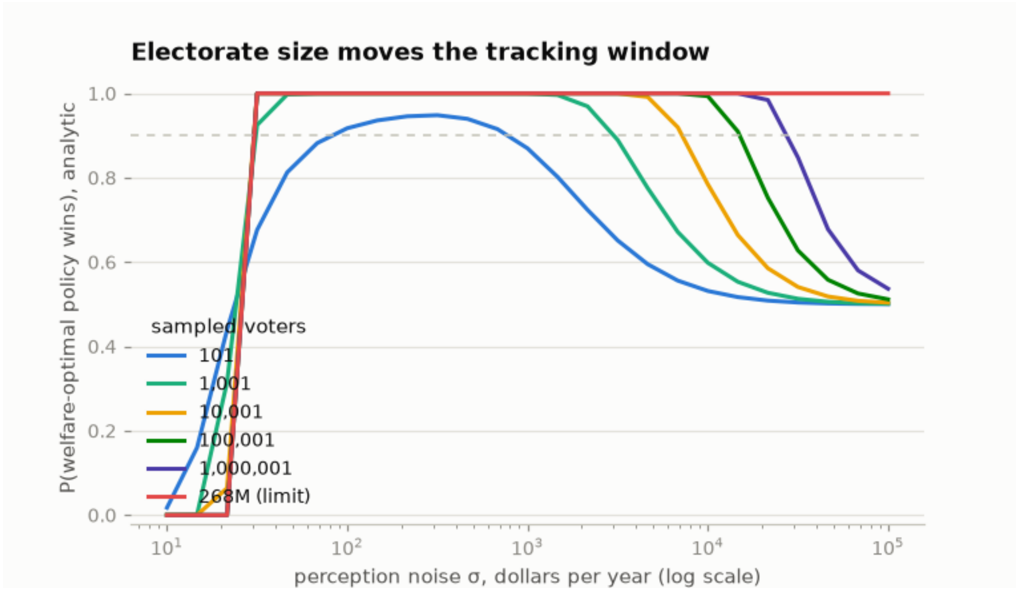

Figure 3: Analytic tracking probability by electorate size, per-capita financing, against noise σ (log dollars). On this axis the knife-edge is the fixed left wall near $\sigma \approx \$25$; the right slope, where noise finally drowns the high-stakes signal, is the jury-theorem edge that moves with n.

5 Endogenous platforms

The results so far grade elections against two given policies. This section makes the platforms choices and asks what electoral competition enacts.

The position game. A platform is a pair of intensities $(\alpha, \beta) \in [0,1]^2$ on the two measured incidence vectors: α scales the credit increment ($\alpha = 1$ is the full \$2,200→\$3,200 change), β scales the rate-cap depth. Positions produce gross household deltas $\alpha\, \delta_{iA} + \beta\, \delta_{iB}$, financed per capita; because the levy is linear in the deltas, *net* stakes are exactly linear in (α, β), so the whole space runs on the two committed impact columns and the status quo $(0, 0)$ is a position. Within-family linearity is itself measured, not assumed: two additional engine runs at the midpoint of each family price it. The rate axis is linear to 0.13% of totals (99.95% of households within \$1); the credit axis carries real phase-in convexity — 5.0% of totals, 96% of households within \$1, mean absolute error \$9 — and the grid stays inside the validated range.

Candidates. Each candidate is a household from the data with an objective over enacted positions, in dollars per year:

$$U_c(\alpha, \beta) \;=\; s_c \cdot \Delta_c(\alpha, \beta) \;+\; (1 - s_c) \cdot E(\alpha, \beta), \tag{2}$$

where Δ_c is the candidate's own household net delta, E is the position's change in equally-distributed-equivalent (EDE) household income under the candidate's inequality aversion (Atkinson, 1970), and s_c is the selfish weight. The EDE is an increasing transform of the social welfare function, so it ranks positions identically while being denominated in dollars — own stakes and societal value mix with no normalization, and $s_c = 0.5$ trades a dollar of the candidate's own for a dollar of society-wide EDE income per household. An optional office rent adds a fixed

Candidates	$\sigma = \$0$	\$200	\$1,000	\$10,000	\$30,000
Office-seekers (rent only)	SQ	SQ	SQ	SQ	SQ
Both societal ($\eta = 1$)	SQ	SQ	SQ	SQ	SQ
Both selfish	SQ	.01/0	.19/.05	.65/.25	.55/.45
(proposals)	—	(.05/0)	(.35/.10)	(1.0/.70)	(1.0/1.0)

Table 2: Mean enacted position across pure Nash equilibria, written α/β (credit and rate-cap intensity); SQ is the status quo $(0, 0)$. Parenthesized rows give the two candidates' proposals from best-response dynamics. The welfare-optimal position under $\eta = 1$ is the status quo: both financed endpoint policies score below it.

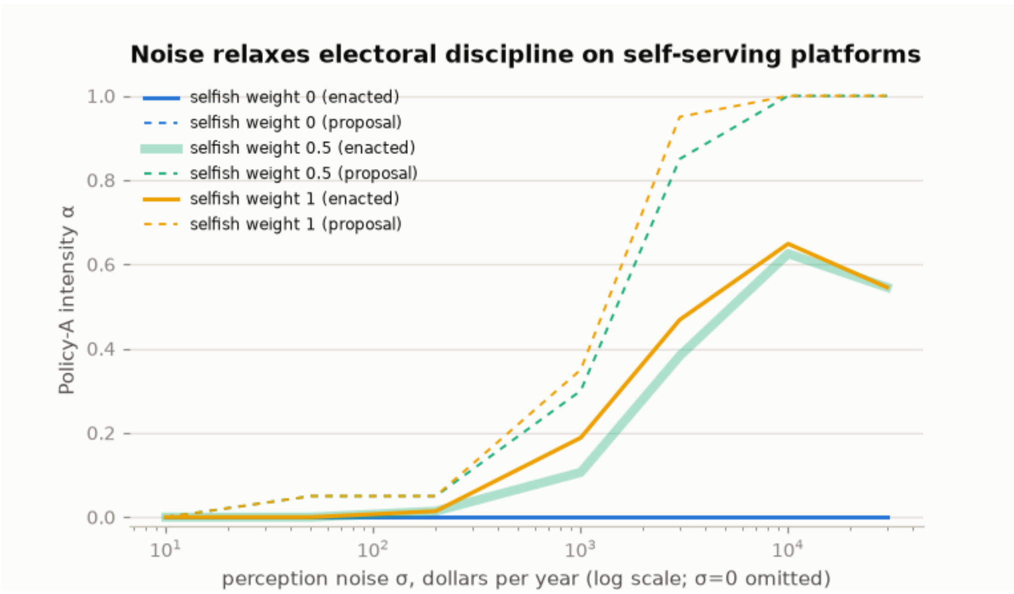

Figure 4: Equilibrium credit intensity against perception noise, by the candidates' selfish weight. Solid lines are win-probability-weighted enacted intensity; dashed lines are the credit candidate's proposal. Perfect information pins every equilibrium to the status quo; noise lets self-serving programs through.

dollar win bonus. Expected utility is $P(\text{win}) \cdot U_c(\text{own position}) + (1 - P(\text{win})) \cdot U_c(\text{opponent's})$, with $P(\text{win})$ from the same probit-and-normal-approximation machinery as equation (1) at 10,001 voters. For self-interested voters the win probability depends only on the position *difference*, so one $(2G-1)^2$ table prices all G^4 profile comparisons on the 21×21 grid and exhaustive pure-Nash enumeration is exact and cheap. The headline candidates are selected by measured stake and asserted against the committed deltas: a two-adult, three-child household at the median income of such households (\$109,693; gross credit stake +\$1,133), and the household at the weighted median income of the rate cap's gross winners (\$578,540, childless; gross stake +\$4,655). An income-percentile proxy fails at this step — the childless 95th percentile sits below the capped brackets and would *lose* from the rate cap.

Three regularities organize Table 2 and Figure 4. First, **perfect information disciplines platforms to the status quo**. At $\sigma \leq \$10$ every equilibrium enacts $(0, 0)$ whatever the candidates want: the same trivial-stakes mass that created the fixed-platform knife-edge now punishes any proposer of a levy, so any program hands the opponent a cheaper-platform win and competition converges on proposing nothing. Under this welfare metric that is the optimum. Perfect information produces the welfare-best outcome through competition — inverting Section 4.2, where the same information regime made the fixed-platform outcome welfare-independent. The inversion is the paper's sharpest comparative static: whether accurate voters help depends on whether the platforms are given or chosen.

Second, **noise relaxes the discipline, and self-interest fills the gap**. As σ grows both candidates escalate their own programs — the credit candidate from a doomed $\alpha = 0.05$ proposal at σ = \$50 (enacted: still zero) to the full program by σ = \$10,000; the rate-cap candidate to the full cap by σ = \$30,000. Halving the selfish weight changes little (enacted α 0.11 versus 0.19 at σ = \$1,000, converging by \$30,000): candidates' own stakes are thousands of dollars while full-intensity societal values are −\$169 and −\$379 of EDE per household, so only near-total societal weight changes candidate behavior. Under fixed platforms, moderate noise rescued welfare tracking; with endogenous platforms the same noise is what lets welfare-negative programs through. Whether misperception helps depends on who sets the agenda.

Third, **competition prices constituency breadth, and noise erodes the price**. The credit program's gross winners are a fifth of adults; the rate cap's are 3%. At σ = \$1,000 that breadth gap keeps enacted intensity four-to-one in the credit's favor; by σ = \$30,000 both candidates run their full programs and the election is a near-coin flip (win probability 0.545 versus 0.455). Enough noise erases the electorate's ability to distinguish a broad program from a narrow one — the breadth filter is a head-count, indifferent to the welfare metric, and it only operates while voters can see their stakes.

6 Heterogeneous preferences

Baseline voters maximize perceived own income. This section gives them the candidates' utility family: voter i of type t values policy j at $s_t\hat{\delta}_{ij} + (1-s_t)\hat{E}_{tj}$, where $\hat{\delta}_{ij}$ is the perceived own stake of equation (1)'s model, $\hat{E}_{tj}$ is the policy's EDE change under the type's inequality aversion η_t perceived with type-specific noise and bias, and s_t is the type's selfish weight. Both terms are dollars. A population is a mixture of types — heterogeneous in s, η, and information quality — and the mixture is exact in the analytic machinery (types enter as weights, not draws). Every voter margin remains normal, so the closed forms of Section 2 extend, and the equilibrium tables of Section 5 take arbitrary mixtures.

A sliver of societal motive cures the knife-edge. The knife-edge ruled because the trivial-stakes mass voted a −\$4.10 levy margin. The societal signal is fifty times larger: the two policies' EDE values differ by \$209.77 per household (−\$169.29 versus −\$379.06, $\eta = 1$). At $\sigma = 0$, tracking flips from 0 to 1 as the homogeneous selfish weight falls through $s^* = 0.9623$: voters who place under 4% of their utility weight on society still vote as if they could see the welfare ranking. Higher weights buy robustness at the noisy end instead — at σ = \$30,000, where baseline tracking has decayed to 0.546, tracking is 0.694 at $s = 0.5$, 0.902 at $s = 0.25$, and 1.0 for sociotropic voters (Figure 5). The two informational channels are not symmetric: own-stake noise cancels only through the plurality aggregate, while an accurately perceived societal signal is common to every voter who weights it.

Value pluralism is its own distortion channel. Sociotropic voting removes the misperception distortions and introduces one that no perception survey can close: fully informed voters

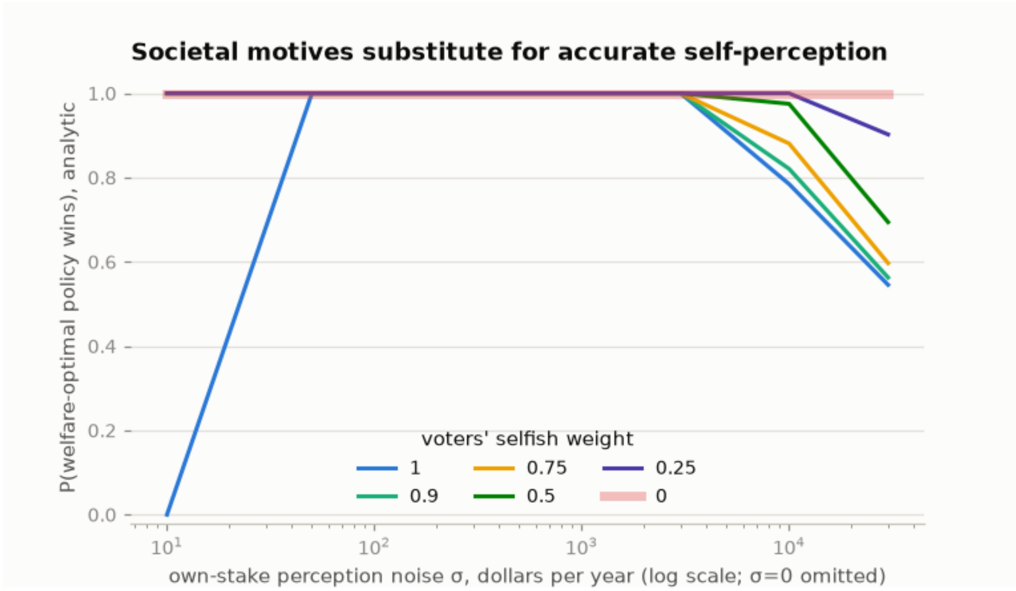

Figure 5: Analytic tracking against own-stake noise by the voters' selfish weight (informed societal component, 10,001 voters). Societal motives substitute for accurate self-perception.

disagree because they hold different welfare functions. On these policies the EDE ranking flips between $\eta = 2$ and $\eta = 2.5$ — at 2.5 the rate cap edges the credit by all of \$2.88 per household ($-\$696.74$ versus $-\$699.62$). Split an all-informed sociotropic electorate between $\eta = 1$ and $\eta = 2.5$ types and the outcome follows the majority lens; graded against either welfare function, every majority composition is simultaneously "tracked" under one and "failed" under the other, and the 50/50 split is a coin flip under both.

A sociotropic minority disciplines candidates. The escalation of Section 5 assumed every voter is a noisy self-interested one. Mixing in a share q of informed sociotropic voters ($s = 0$, $\eta = 1$) against the same purely selfish candidates dismantles it (Table 3, Figure 6). At $q = 0.10$ every noise level has pure equilibria again and all of them enact the status quo: the bloc votes deterministically against whichever candidate proposes the more welfare-negative program, and a tenth of the electorate outweighs the selfish margin any program can buy among its beneficiaries. The transition is not smooth: at $q = 0.02$–0.05 and high σ the pure-equilibrium set is *empty* — proposing a program wins the noisy selfish margin until the opponent undercuts and takes the sociotropic bloc, which restores the incentive to propose, and best responses cycle with proposals reaching up to full intensity inside the cycle. The bloc also degrades gracefully: sociotropic voters who misread the societal value with \$500 of noise per household (against a signal peaking near \$380) still cut enacted intensity at 30% share from 0.65 to 0.11, while an electorate that is uniformly *half* selfish — no informed bloc at all — cuts it only from 0.63 to 0.44 on a matched grid. A small bloc that votes the societal signal deterministically disciplines far more than diffuse partial altruism, which merely shrinks every voter's selfish margin.

Sociotropic share q	$\sigma = \$1{,}000$	\$10,000	\$30,000
0 (baseline)	0.19	0.65	0.55
0.02	cycling	cycling	cycling
0.05	0.00	cycling	cycling
0.10–0.50	0.00	0.00	0.00

Table 3: Mean enacted credit intensity across pure Nash equilibria, purely selfish candidates. "Cycling" marks empty pure-equilibrium sets, where best-response dynamics orbit instead; 0.00 is zero to numerical precision.

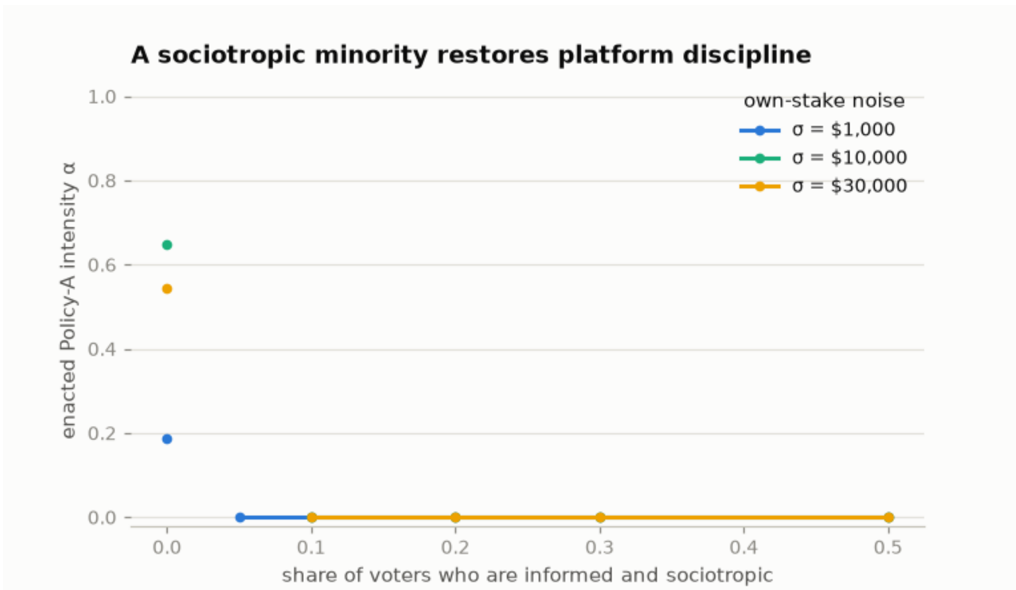

Figure 6: Enacted credit intensity against the informed-sociotropic voter share, by own-stake noise. Gaps are the cycling regime of Table 3.

7 Robustness and limitations

The welfare ranking is stable across $\eta \in \{0.5, 1, 2\}$ and under per-capita and square-root equivalence scales, but flips to Policy B by $\eta = 2.5$ (holding at the higher tested values), where the \$1,000 income floor's censoring of the poorest households' levies dominates; the ranking is a modeling choice, reported as such, and Section 6 converts the sensitivity into a mechanism. Both financing rules hand the no-stakes bloc its signed sliver; financing choices that zero it, or deficit financing that hides it, restore monotone tracking. Errors are independent across voters — including adults in the same household — and across policies; household-correlated error is not simulated here and could change the levy bloc's effective behavior.

The endogenous layers carry their own scope. The position game is two candidates, one shot, pure strategies on a grid, in the two-dimensional space the measured incidence vectors span; mixed strategies in the cycling regime are not computed (best-response dynamics are reported instead), and entry, primaries, dynamics, and commitment problems are out of scope. Voter types mix independently of stakes and demographics; the societal signal a voter perceives is the EDE change of the electorate actually voting; and the societal noise, bias, and η distributions are labeled assumptions with nothing measured behind them — the layer exists so that survey estimates of sociotropic perception can drop in beside the own-stake ones.

The binding limitation is that perception is assumed. The linear-Gaussian family is exactly what a survey regression of perceived on true impacts would estimate — attenuation, bias, and residual dispersion, by demographic cell — and the model's perception interface accepts a fitted empirical distribution without modification. Measured tax-policy knowledge, beliefs, and perceived distributional effects exist in the literature (Bartels, 2005; Stantcheva, 2021), but not the engine-computed, own-household incidence beliefs this model consumes; eliciting it is the natural next step, and its answer is a single point on the axes swept here.

8 Conclusion

On measured household stakes, the relationship between voter accuracy and welfare tracking inverts the toy intuition at both ends: elections track almost perfectly across two orders of magnitude of misperception, fail on a welfare-irrelevant residual at perfect information unless the trivial-stakes majority goes silent, scale their thresholds with electorate size, and trade a thousand dollars of noise tolerance for a two-hundred-dollar bias vulnerability. Making the platforms choices flips the verdict on information: perfect information disciplines competing candidates to the status quo — here the welfare optimum — while the same noise that rescued fixed-platform tracking is what lets self-serving programs through, eroding even the electoral advantage of broad constituencies over narrow ones. And the cheapest repairs live in preferences, not information: a few percent of utility weight on society cures the knife-edge, and a tenth of the electorate voting an informed societal signal disciplines purely selfish candidates at any noise level, leaving value pluralism — informed voters with different inequality aversions — as the distortion no measurement closes. The mechanisms are old; the denominations are new. Whether real electorates sit on the tracking side of these numbers — how accurately voters see their own stakes, how much weight they place on society's, and through which welfare lens — is now a measurement question.

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